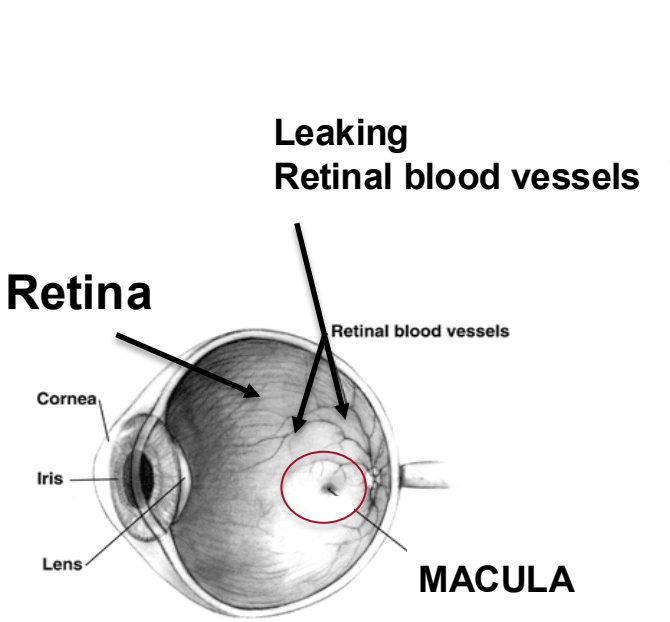
Macular edema

Ashok Puri, **MBBS, MS**

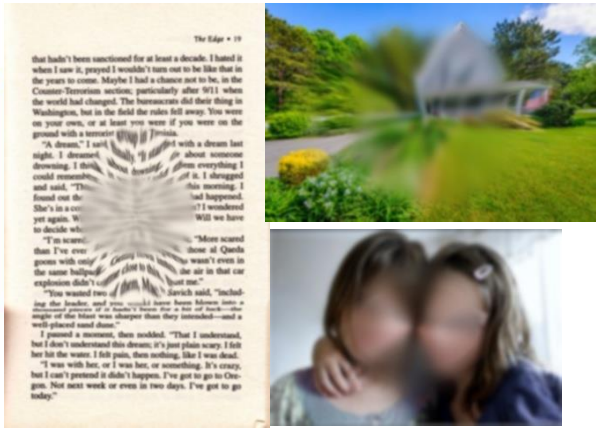
Assistant Professor
Department of Ophthalmology,
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Omaha NE
Email: apuri@unmc.edu

University of Nebraska Medical Center

Macular edema: swelling of the retina of the eye







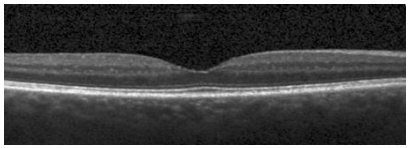
Leading cause of vision loss working age adults *(1.5M U.S.)*

- Diabetes
- Retinal vein occlusion
- Age related macular degeneration

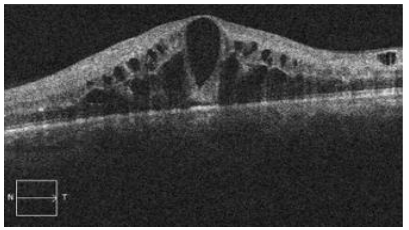
Macular Edema: Diagnosis



Retinal optical coherence tomography (OCT)



Normal scan



Macular edema

No cure, only control

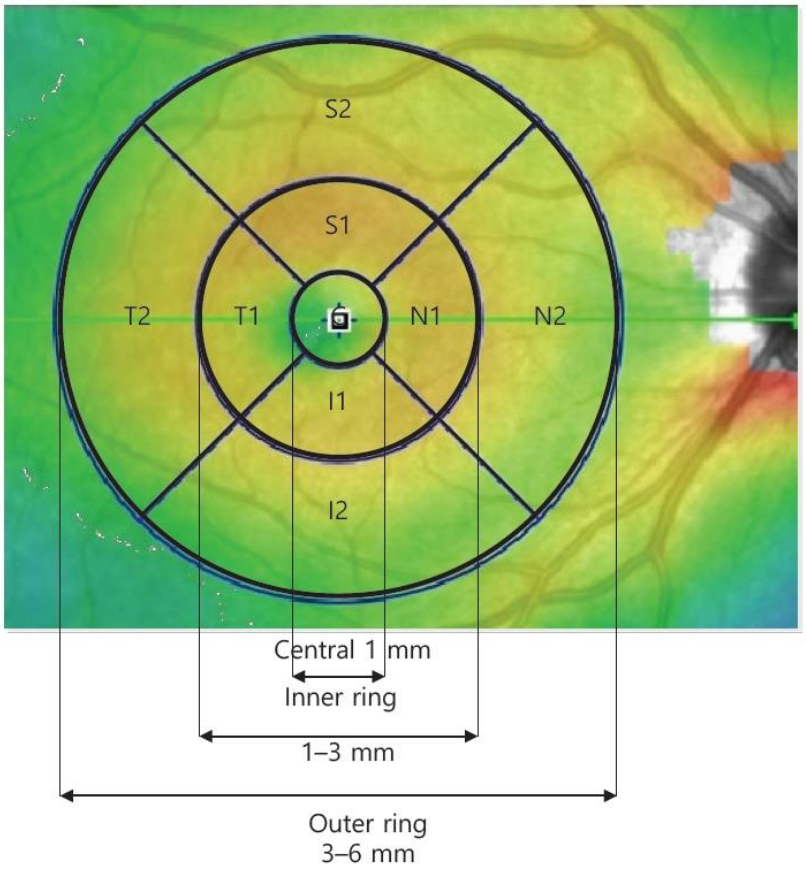
- ***Multiple anti-VEGF injections***
- ***Costly: $2000/injection***



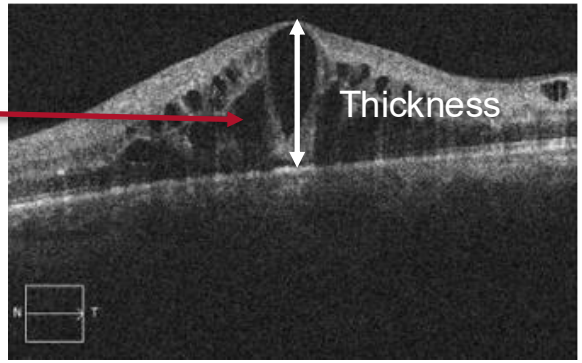
Blanket monthly injections for all
Attempts to personalize:
TREX – Reactive approach

Data-driven Pro-active approach

Macular Edema: Biomarkers



Fluid Collection



Central subfield thickness (CST)

Intra retinal fluid (IRF) volume

+

Qualitative biomarkers

OLIVES dataset

Monthly – 12 eyes

24-27 visits

49 scans from top to bottom across the 6mm^2 macula

TREX – 22 eyes

variable visits

49 scans from top to bottom across the 6mm^2 macula

Data

OCT images

CST

Vision

Presence of qualitative biomarkers

Injections:

Monthly – 1 injection each month

Continued suppression

TREX – monthly injections till CST ≤ 325 microns

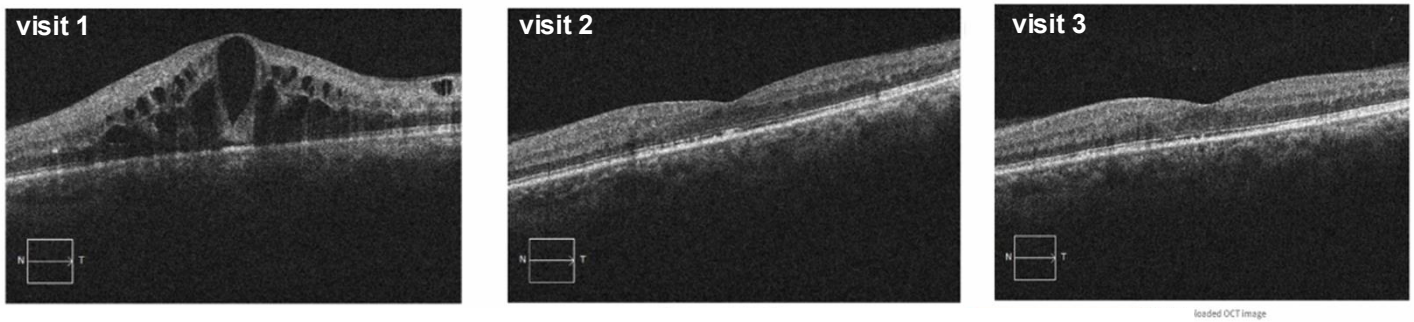
No injections till CST≥ 325 microns

20% ↑CST from nadir

Retinal Texture & Pattern

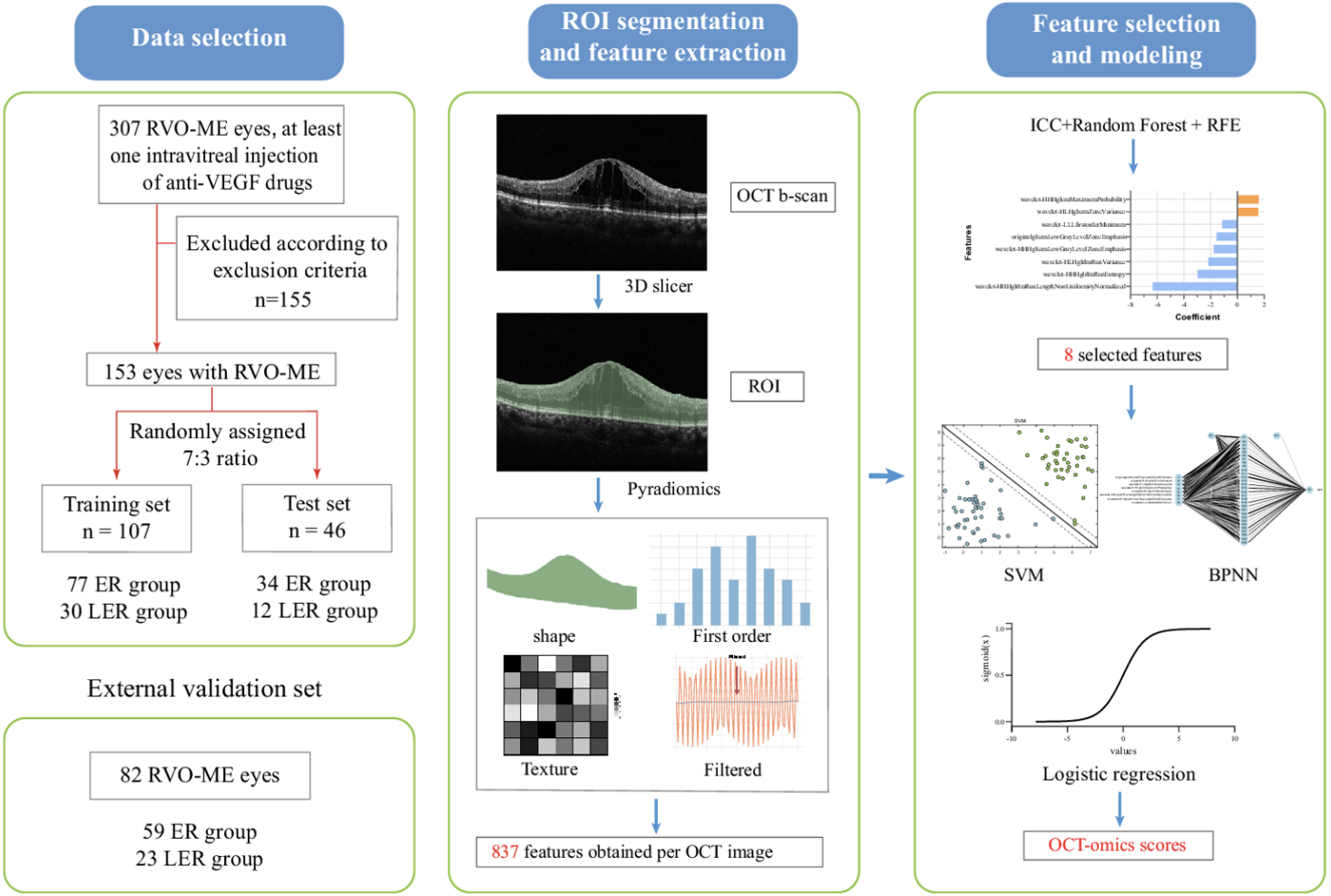
(*The Lead Indicator)*

Scans of a patient on monthly injections



Radiomics

Radiomics



Radiomics Analysis Based on Optical Coherence Tomography to Prognose the Efficacy of Anti-VEGF Therapy of Retinal Vein Occlusion-Related Macular Edema

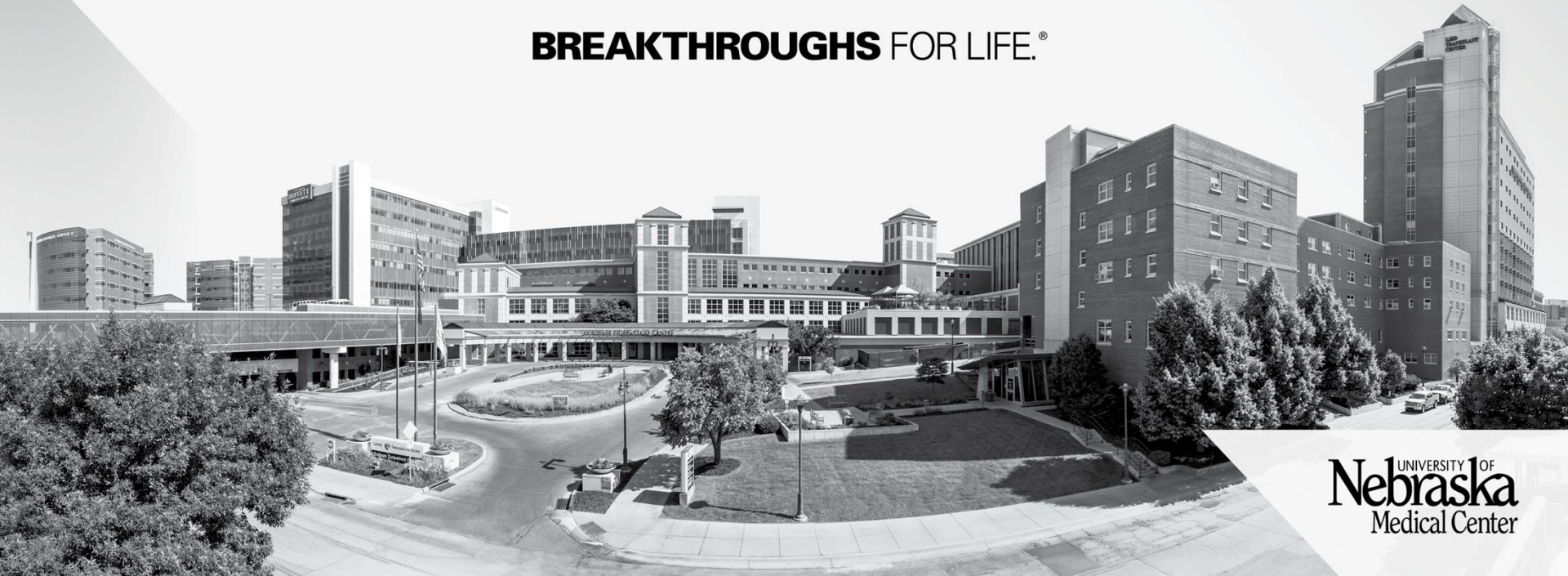
Investigative Ophthalmology & Visual Science April 2025, Vol.66, 74. doi: https://doi.org/10.1167/iovs.66.4.74

Purpose

- **Severity scoring**
- **Rate of change/visits**
- **Treatment-response analysis & prediction (Cox, t+1…)**
- **Trajectory forecasting** (future CST/VA/recurrence)
- **Treatment-timing Decision** (R/Y/G)

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